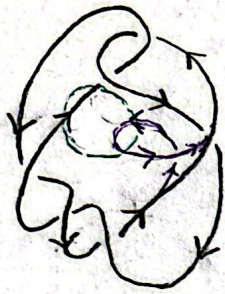


[LEMMA: RIBBON $\Rightarrow$ SLICE]

6_1 :

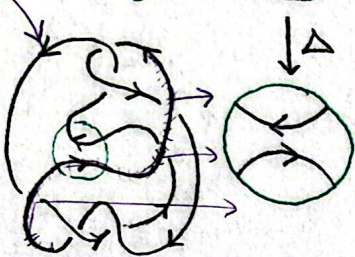


• ISOTOPY

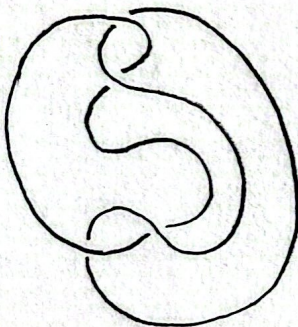
SADDLE



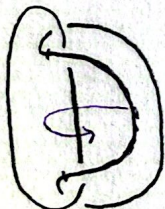
(Δ map)
Two COMPONENTS



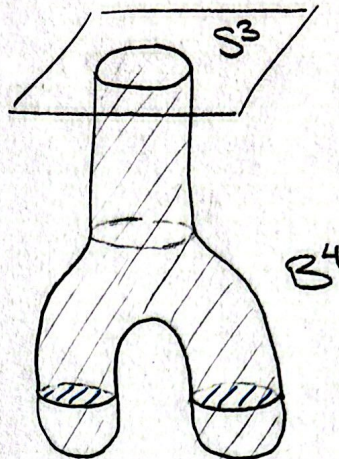
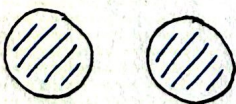
ISOTOPY



ISOTOPY

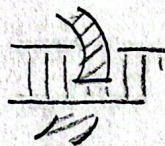


ISOTOPY



SOLID DISK $\Rightarrow$ SLICE?

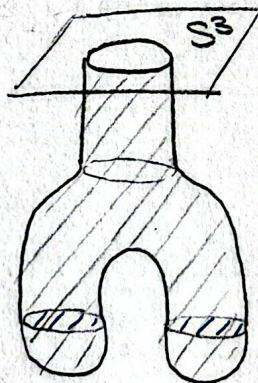
DEF: RIBBON: A knot is ribbon if it bounds a disk that crosses through itself only in arcs. In particular, $[K \# -K]$ is ribbon.



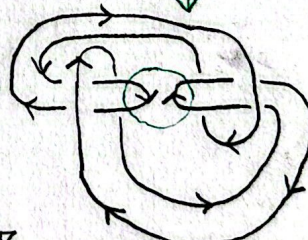
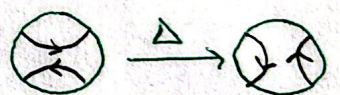
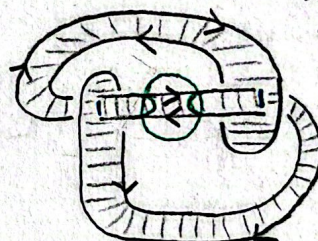
EXAMPLE:

ARG: "Push INTO B^4 "

not convincing?
can also see via a saddle?



Again, a solid disk? $\Rightarrow$ SLICE?



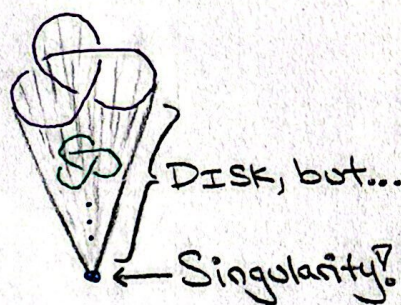
ISOTOPY



Another way to represent 6_1

DEF: SLICE: $K \subset S^3$ is slice $\iff K$ bounds a smooth disk in B^4

$\hookrightarrow$ need smooth, else note every knot bounds a cone in B^4 , where pt is a singularity
 $\therefore$ think: how in calc smooth $\Rightarrow$ nice, infin. diff.!

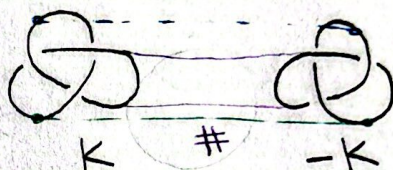
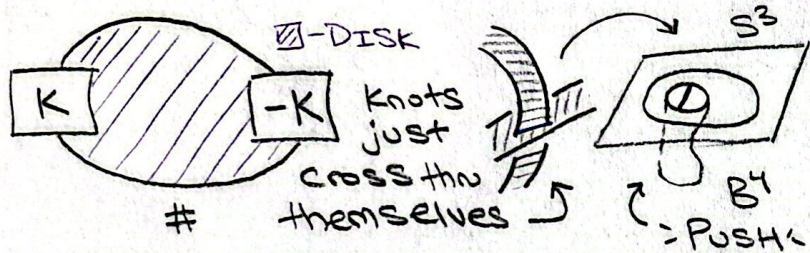


WHY SLICE IMPORTANT? $\therefore$ EQUIVALENCE RELATION?

DEF: CONCORDANT: knots K, J are concordant ($K \simeq J$)

if $K \# -J$ is slice? Claim: Concordance is an equiv. rel. on knots?

REFLECTIVE: $K \simeq K$ WTS $K \# -K$ is slice.

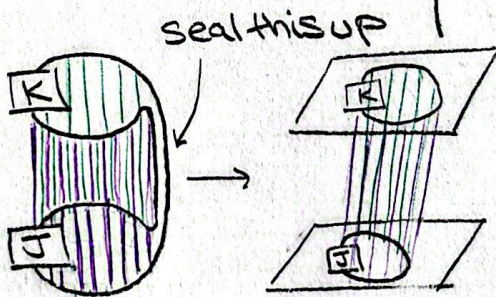
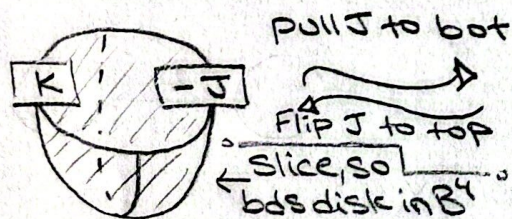


SYMMETRIC: $K \simeq J \Rightarrow J \simeq K$ Assume $K \# -J$ slice. WTS $-J \# K$ slice.

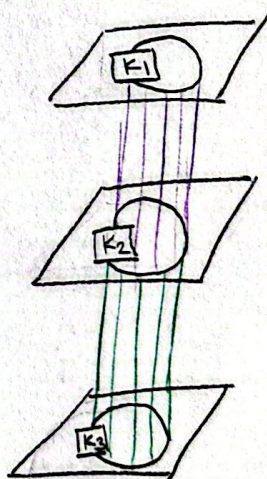
$-J \# K = K \# -J \leftarrow$ indeed is given slice
 $\uparrow$ commutative

TRANSITIVE: WTS $K_1 \simeq K_2, K_2 \simeq K_3 \Rightarrow K_1 \simeq K_3$

EQUIV DEF of CONCORDANCE:



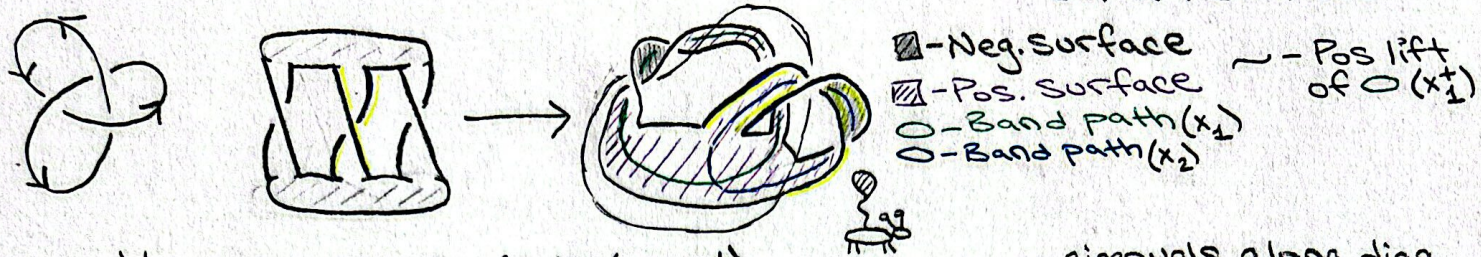
TRANSITIVE:



$\Rightarrow$ Mirror of a knot is its inverse under concordance?

$\mathcal{C}$ is the KNOT CONCORDANCE GROUP?

SEIFERT SURFACES Any knot bounds orientable surface of disks & bands



SEIFERT MATRIX: each entry is $lk(x_i, x_j^+)$

$$V = \begin{matrix} x_1^+ & x_2^+ \\ x_1^+ & \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ x_2^+ & \end{matrix} \rightsquigarrow V+V^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \rightarrow \text{Symmetric } V+V^T \Rightarrow \text{diagonalizable}$$

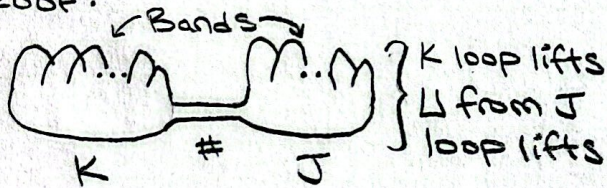
SIGNATURE OF MATRIX: #pos eigenvals - #neg eigenvals

$K \rightsquigarrow$ Seifert Surface $\rightsquigarrow$ Seifert Matrix $\rightsquigarrow \sigma(K)$ signature

PROPERTIES OF SIGNATURE:

$$\sigma(K \# J) = \sigma(K) + \sigma(J)$$

PROOF:



$$V_{K \# J} = \begin{pmatrix} V_K & 0 \\ 0 & V_J \end{pmatrix} \Rightarrow V+V^T = \begin{pmatrix} V_K+V_K^T & 0 \\ 0 & V_J+V_J^T \end{pmatrix}$$

$$\Rightarrow D_{K \# J} = \begin{pmatrix} D_K & 0 \\ 0 & D_J \end{pmatrix} \left\{ \begin{array}{l} \text{diagonal} \\ \text{breaks down} \end{array} \right.$$

$$\Rightarrow \sigma(K \# J) = \sigma(K) + \sigma(J).$$

$$\sigma(K) = -\sigma(-K)$$

all crossings flipped $\Rightarrow$ linking #'s all flip

$$\Rightarrow V_{-K} = -V_K \Rightarrow D_{-K} = -D_K \Rightarrow \sigma(K) = -\sigma(-K).$$

K SLICE $\Rightarrow B^4$

By (*) we use thanos topology $\Rightarrow$ the small vertical circles can shrink down and disappear.

Note, in B^4 so interior is actually $S^3 \neq$

$$\Rightarrow \exists V_K = \begin{pmatrix} 0 & A \\ B & C \end{pmatrix} \begin{matrix} g \\ g \end{matrix}$$

where g is genus(K)

$$\Delta[\cdot] \Rightarrow \approx \begin{pmatrix} 0 & D \\ D^T & 0 \end{pmatrix}$$

$$\Rightarrow V_K+V_K^T = \begin{pmatrix} 0 & A+B^T \\ A+B & C+C^T \end{pmatrix} \Rightarrow \begin{matrix} g \times 1 \text{ eigenvals} \\ g \times (-1) \text{ eigenvals} \end{matrix}$$

LEMMA: K Slice $\Rightarrow \sigma(K) = 0 \rightarrow$ PROOF

COR: Suppose J has Seifert surfaces

F_1, F_2 . Then $\sigma_{F_1}(J) = \sigma_{F_2}(J)$ well-def. PROOF: $\sigma_{F_1}(J) - \sigma_{F_2}(J) = \sigma_{F_1}(J) + \sigma_{F_2}(-J)$

COR: Suppose J1 concordant to J2. $\Rightarrow \sigma_{F_1 \# F_2}(J \# -J) = 0 \Rightarrow \sigma_{F_1}(J) = \sigma_{F_2}(J)$

$\Rightarrow \sigma(J1) = \sigma(J2)$. PROOF:

$$J1 \# -J2 \text{ slice} \Rightarrow 0 = \sigma(J1 \# -J2)$$

$$= \sigma(J1) + \sigma(-J2)$$

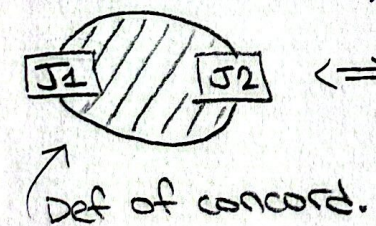
$$= \sigma(J1) - \sigma(J2)$$

$$\Rightarrow \sigma(J1) = \sigma(J2).$$

NOTE: CONVERSE, i.e., $\sigma(K) = 0 \Rightarrow K$ slice IS NOT TRUE! EX. (4,1), CONWAY K

Alexander poly: $V^T - tV$. So here, we specialize with $t = -1$

$\Delta(-1)$ spec. case



Def of concord.